MILITARY SPECIFICATION
RESTORER, PULSE FORM TD-206B/G

This amendment forms a part of Military Specification MIL-R-55602A(EL), dated 30 June 1976, and is approved for use by the Communications-Electronics Command, Department of the Army and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

2.1, DRAWINGS, Electronic Command

delete "SM-E-704762" and substitute "SM-E-955736".

add: "SM-D-955737 Cable Signal Test Unit Assembly Schematic Diagram."

PAGE 6

3.3, line 4, add: "(see 4.4)".

3.4.1.1, Delete in its entirety and substitute the following:

"Each restorer shall receive and process timing rates from 2.304MHz-0.003% to 2.304MHz +.003%. A 144KHz square wave input as shown in Figure 9 shall not induce jitter in excess of 30ns peak to peak total when measured at any point in a 40 mile cable system after the first restorer, using a Psuedo Random Pattern Generator (PRPG) having a minimum of 13 stages (see 4.9.14)"

PAGE 7

3.4.1.2, change as shown:

From:		To:	
Carrier Frequency	Peak Signal to RMS Carrier Ration	Carrier Frequency	Peak Signal to RMS Carrier Ratio
300 KHz	-10	300 KHz	-7
500 KHz	+1	500 KHz	+3
800 KHz	+9	800 KHz	+11
1152 KHz	+15	1152 KHz	+17
2304 KHz	+25	2304 KHz	+25
3000 KHz	+24	3000 KHz	+26
4500 KHz	+18	4500 KHz	+24

FSC 5805

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3.4.1.2, line 6: Delete "minimum".

3.4.1.3, delete "180 to 230 ns" and substitute "190 to 250 ns".

PAGE 8

3.4.4, replace the first sentence with the following:

"With a 2.304 MHz +.001% PCM signal at 6.6 mV rms, set at 38 mA + 2%, the voltage drop across the restorer shall be 7.250 +0.50 Vdc and the voltage at the input of the restorer shall be 21.24 +0.50 Vdc.

3.4.5, change title to "Electrical current surge and voltage transient protection".

PAGE 11

3.7.6, change title to "Leakage (immersion)".

PAGE 12

3.11, line 6, add: "(See 4.13)".

PAGE 13

3.16, change title to "Bounce preconditioning".

PAGE 14

3.20, line 4: delete "paragraph" and substitute "paragraphs".

PAGE 15

4.2.c, Delete in its entirety and substitute:

"c. Quality conformance inspection of equipment before packaging (See 4.5)".

4.2, Add new subparagraph as follows:

"d. Packaging inspection (See 4.16)".

PAGE 16

4.5, change title to:

"Quality conformance inspection of equipment before packaging".

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PAGE 17

TABLE I., After title add: "(see Notes A and B)".
TABLE I., Inspections column: delete "Current surge and voltage transient" and substitute "Electrical current surge and voltage transient protection".

TABLE I., Inspections column: change "Immersion" to "Leakage (immersion)".

PAGE 19

b., line 8; delete "pruchase" and substitute "purchase".

c., line 4; delete "3.11" and substitute "3.16, 3.11".

PAGE 20

4.5.1.1, Delete in its entirety and substitute the following:

Order of Inspection with Group A "Group A inspection shall be performed in the order shown in Table II.".

TABLE II, Req. Para. column: delete "3.4.12" and substitute "3.4.1.2".

TABLE II, Req. Para column for "Operational (output signal) ":delete "3.4.1.2 3.4.2. 3.4.2.1".

PAGE 21

4.5.2, line 2: delete "in" and substitute "on".

TABLE III, AQL column: delete "groups" and substitute "group".

4.5.2, delete the first line in Table III which reads "Timing Recovery; 3.4.1.1; 4; 4.9.8".

4.5.2.2, in title: delete "with" and substitute "within".

PAGE 22

4.5.3.1.2, line 5: delete "EMI" and substitute "EMC".

PAGE 23

TABLE IV., Inspection column: delete "Current surge and voltage transient" and substitute 'Electrical current surge and voltage transient protection".

TABLE IV., Inspection column: delete "Immersion" and substitute "Leakage (immersion)".

TABLE IV., Requirement Paragraph column for Shock (drop): Change "3.7.4.3" to "3.7.4.4".

PAGE 24

4.5.5, lines 3 and 4: Delete "and Group B".

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PAGE 25

4.8.1.1, Delete in its entirety and substitute the following: "Pretest data required is specified in 4.9.7.".

PAGE 27

4.8.8, change title to: "Leakage (immersion) test.".

PAGE 28

4.8.11, line 2: delete "Method 510" and substitute "Method 510.1".

PAGE 29

4.9.4 a., delete "SM-E-704762" and substitute "SM-E-955736".

PAGE 30

4.9.4 b., Delete in its entirety and substitute the following: (b)) Cable Signal Test Circuit (in accordance with SM-D-955737). This circuit is used to test the effect of noise and carrier on the TD-206( )/G. It allows adjustment of the level of the FROM CABL PCM signal from 3.5 to 8.84 milivolts RMS through 1 or 3/4 mile of CX-11230( )/G cable to the next TD-206( )/G input.

4.9.4 c., line 3, delete "four(4)" and substitute "sixteen(16)".
line 5, change "from" to "for".

4.9.4 d., delete "SM-E-704762" and substitute "SM-E-955736".

PAGE 32

4.9.7 f., delete "-0.005%" and substitute "-0.003%"

4.9.7 h., delete "+0.005%" and substitute "+0.003%"

4.9.7 j., line 1: Delete "3.4.1.1".

4.9.8, Delete in its entirety.

PAGE 33

4.9.9 g., Delete in its entirety and substitute the following: "Repeat c. through f. above for 5.3 and 3.5 milivolts RMS at 2.30MHz for 50% "1" pattern".

PAGE 34

4.9.9 j., insert at the beginning of the paragraph: "At a signal level of 8.84 milivolt RMS...".

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4.9.9.k., Delete in its entirety and substitute the following: "Repeat j. above for 3.5 and 5.3 millivolts RMS input to the TD-206( )/G.

4.9.11.b., Delete in its entirety and substitute the following: "Set the input signals at the repeater under test to 6.6mV rms at 2304kHz with the word generator set to a 1010... continuous pattern.".

PAGE 35

Military Specification Number: delete "MIL-R-55620A(EL)" and substitute MIL-R-55602A(EL).

4.9.12, change title to: "Electrical current surge and voltage transient test".

PAGE 36

4.9.14.f., delete "-0.005%" and substitute "-0.003%"

4.9.14.h., delete "+0.005%" and substitute "-0.003%"

PAGE 37

4.9.14.n., delete in its entirety.

4.9.15, Delete in its entirety and substitute the following: "Each circuit card assembly 7A1A shall be tested in accordance with the procedures of the test requirement drawing cited in 3.17.".

PAGE 40

4.14.2, lines 9 and 10: delete "4.9.6a. through 4.9.6e,: and substitute "4.9.6".

PAGE 41

4.14.4, change title to "Sampling for first article reliability inspection"

PAGE 42

4.16, Delete in its entirety and substitute the following:

4.16 Packaging Inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.
b. Quality Conformance Inspection of Packaging.

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4.16.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.16.2 Quality Conformance Inspection of Packaging.

4.16.2.1 Material Inspection All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.16.2.2 Preservation Inspection Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be specified therein.

4.16.2.3 Packing Inspection Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VII. "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same material during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

TABLE VII . PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIA	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance w/specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Veight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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PAGE 43-46

5., Delete in its entirety and substitute the following:

5. PACKAGING

5.1, Preservation. Preservation shall be as specified in MIL-STD-726, Coded as follows:

5.1.1, Level A JR-1-1-ØØ-OO-NS-X-ED-Ø-OO-A

5.1.2, Level B JR-1-1-ØØ-OO-NS-X-ED-Ø-OO-B

5.2, Packing and Marking Packing and Marking shall be in accordance with MIL-E-55585.

PAGE 47

6.2.b, Delete in its entirety and substitute the following:

"b. Level A or B preservation and Packing (see section 5)"

6.2.c, Delete in its entirety and substitute the following:

"c. When First Article Inspection Rough Handling Tests are Not Required"

PAGE 48

6.2, Add the following subparagraph:

"h. When first article packaging inspection test reports require Acquisition Activity approval prior to unit packaging"

PAGE 50

6. Add the following subparagraph:

6.11, Level B Preservation "When Level B preservation is specified, this level or protection will only be used under known favorable conditions during transportation, storage and handling."

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PAGE 51

Figure 1: Delete in its entirety.

PAGE 52

Figure 2: Delete in its entirety and substitute the following: Amended Figure 2.

PAGE 53

Figure 3: Delete in its entirety and substitute the following: Amended Figure 3.

PAGE 54

Figure 4: Delete in its entirety.

PAGE 57

Figure 7: Delete in its entirety and substitute the following: Amended Figure 7.

PAGE 58

Figure 8: Delete "*SME-704762" and substitute the following "*SME-955736.".

PAGE 59

Figure 9: Delete in its entirety and substitute the following: Amended Figure 9.

Custodian:
Army-CR

Preparing Activity:
Army-CR
Project No. 5805-A258

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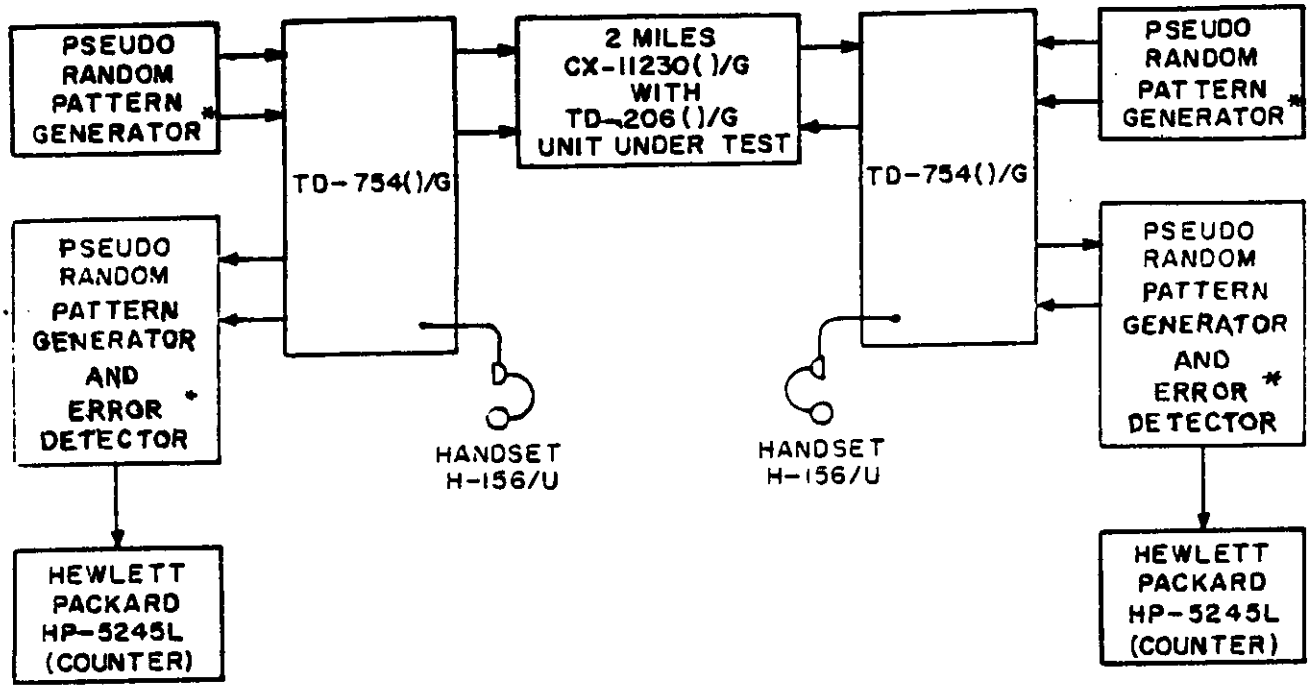
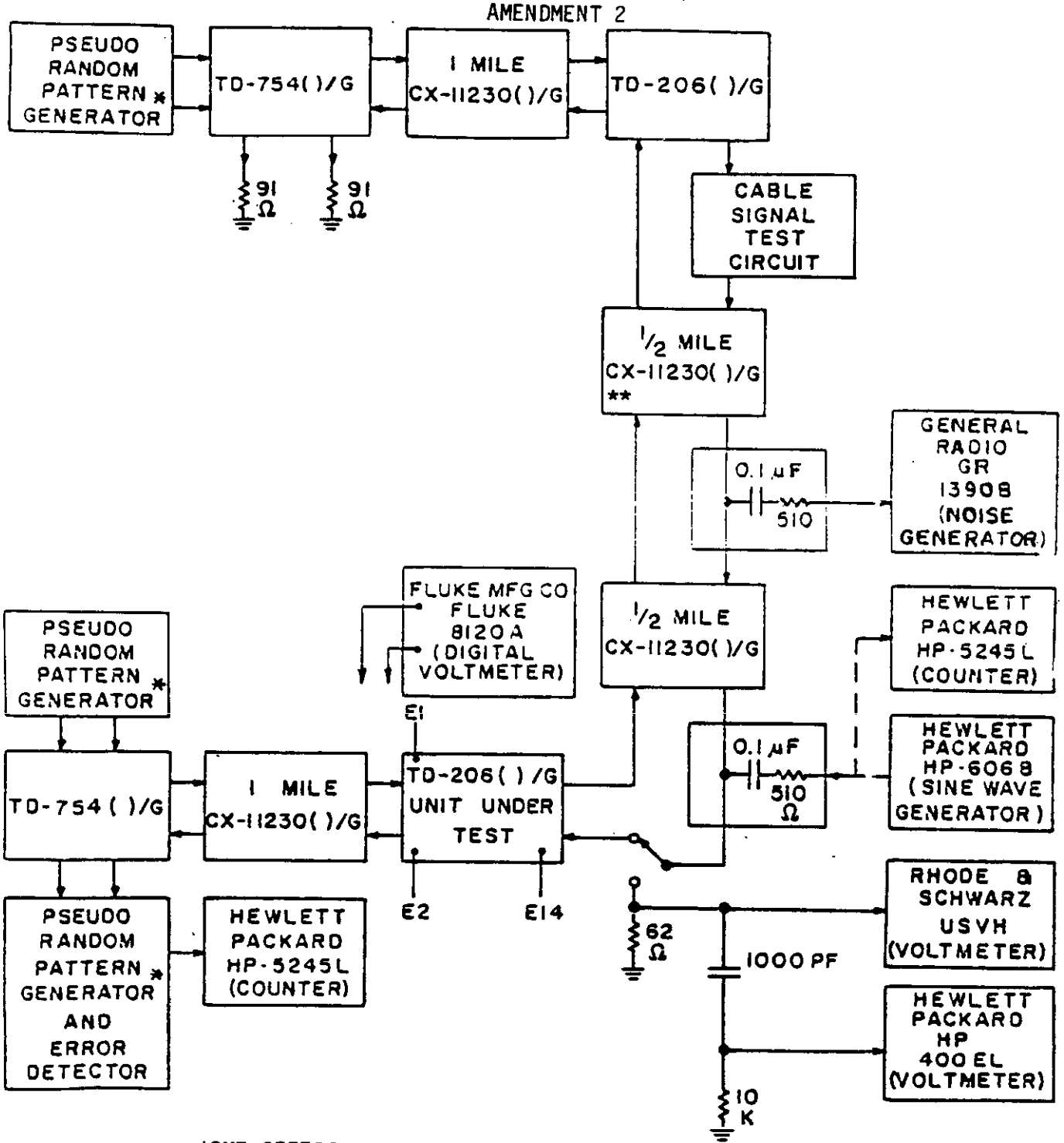


FIG. 2 OUTPUT SIGNAL TEST

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*SME-955736
**MAY BE REDUCED TO ¼ MILE TO ACHIEVE SPECIFIED RMS SIGNAL LEVELS AT TD-206( )/G INPUT

SIGNAL DETECTION TEST & FAULT LOCATION MEASUREMENT TEST

FIG. 5

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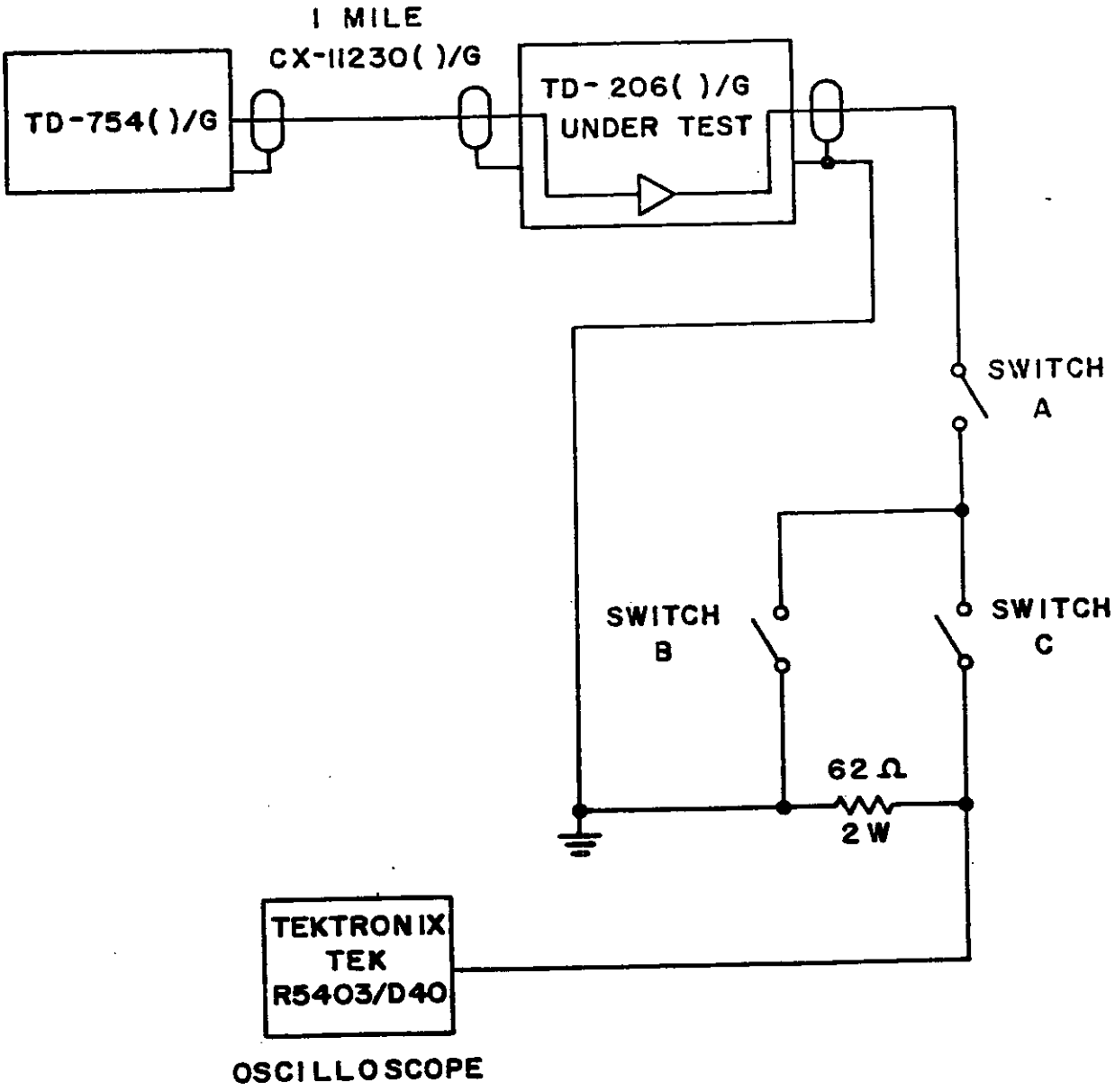
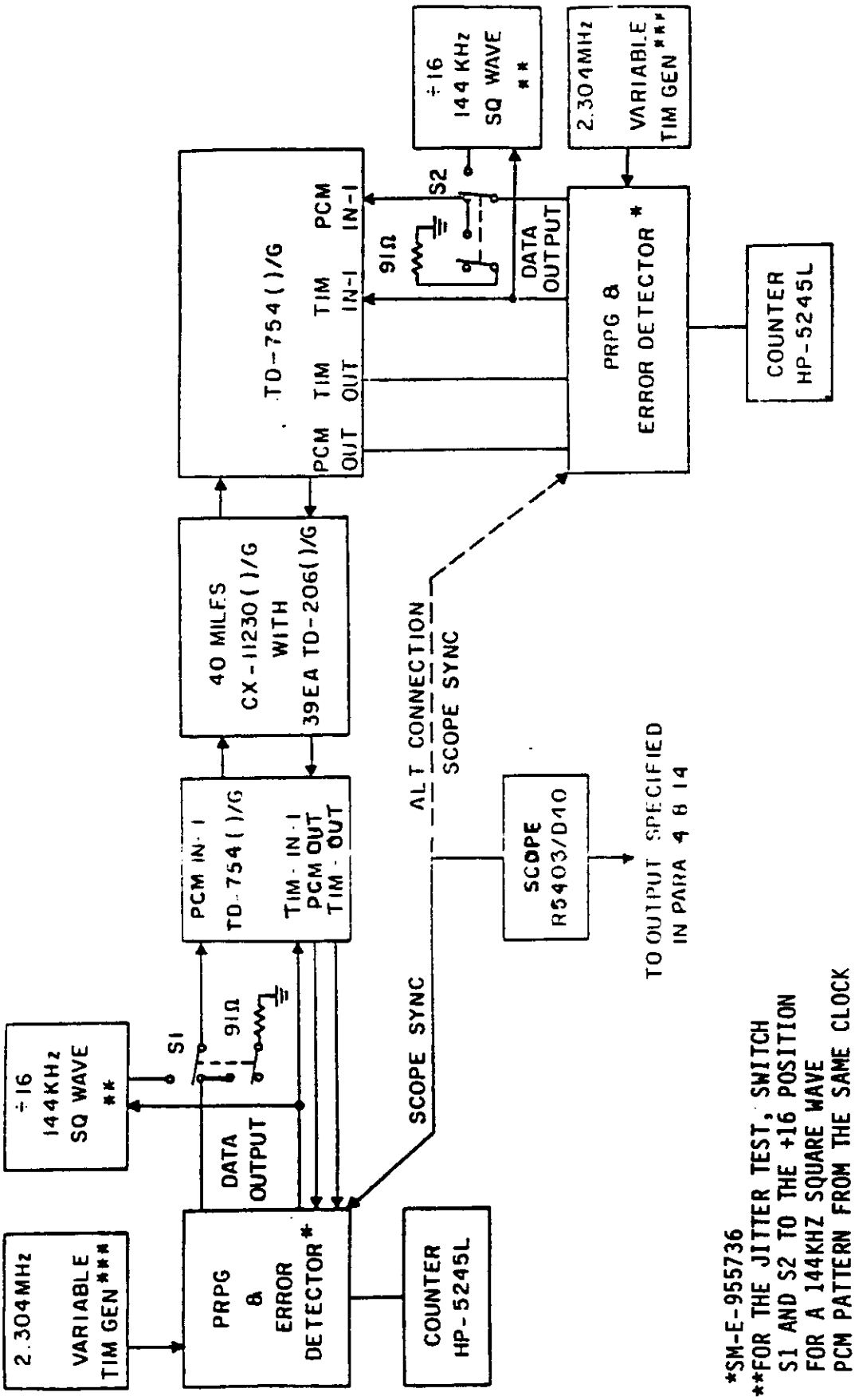


FIG. 7 ELECTRICAL CURRENT SURGE AND VOLTAGE TRANSIENT PROTECTION TEST

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*SM-E-955736
**FOR THE JITTER TEST, SWITCH
S1 AND S2 TO THE +16 POSITION
FOR A 144KHZ SQUARE WAVE
PCM PATTERN FROM THE SAME CLOCK

FIG 9 SYSTEM COMPATIBILITY TEST

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